



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	
Identify	In this incident, the company's internal network and key services used for web design, graphic design, and marketing all stopped working for about two hours. The issue was caused by a DoS attack where a large number of ICMP ping requests flooded the network. After investigating, the team found the main weakness was an unconfigured firewall that allowed this traffic through without any filtering. This outage affected normal business operations because employees couldn't access internal systems or shared resources. The main problem was traced back to missing firewall rules and a lack of traffic controls that should have blocked this kind of flooding.
Protect	To help prevent this from happening again, the company should properly configure its firewall so it can block or limit ICMP traffic and stop suspicious requests from getting through. It should also make sure only trusted traffic is allowed into the network. On top of that, tools like IDS/IPS and network monitoring should stay in place to catch suspicious activity early. Regular checks of firewall settings are also important so misconfigurations don't happen again. Training the IT team can

	help make sure these kinds of setup mistakes are avoided in the future.
Detect	To catch attacks faster next time, the company should rely on network monitoring tools that watch for unusual spikes in traffic, like a sudden flood of ICMP requests. A SIEM system can also help by collecting logs from different systems and flagging anything unusual. An IDS/IPS system should be used to automatically detect and block suspicious traffic patterns. With continuous monitoring, the team can spot attacks early before they cause major disruptions.
Respond	If another attack happens, the first step should be to block the malicious traffic at the firewall and isolate the affected parts of the network. Non-essential services can be temporarily shut down so critical systems stay available. The security team should then review logs to figure out where the attack came from and how it was carried out. Clear communication is also important so employees know what's happening and what steps are being taken. After the attack is under control, firewall and IDS/IPS rules should be updated based on what was learned.
Recover	To recover, the company should slowly bring systems back online once the attack is stopped and everything is stable. During this time, the network should be closely monitored to make sure no more malicious traffic is coming in. After everything is restored, the team should review what happened and see what can be improved. This helps make recovery faster and ensures the same issue is less likely to happen again.

Reflections/Notes:

This incident shows how a simple misconfiguration, like an unprotected firewall rule, can lead to a major network outage. The DoS attack was successful because the network did not have proper filtering in place for ICMP traffic, which allowed the system to be overwhelmed.

It also highlights the importance of layered security. Tools like firewalls, IDS/IPS systems, and network monitoring all worked together after the incident to help detect and reduce the impact of the attack. However, if they had been properly configured beforehand, the attack likely could have been prevented entirely.

Going forward, regular security reviews, better configuration management, and continuous monitoring are key to avoiding similar issues. This incident reinforces the need for proactive security rather than reacting only after systems are affected.

Respond (Future Incident Response Plan)

If something like this happens again, the first thing we would do is contain it quickly by blocking the bad traffic at the firewall and isolating the parts of the network that are affected. This helps stop the attack from spreading and keeps the most important systems running. If needed, we would also temporarily shut down non-essential services to reduce pressure on the network.

To stop the attack, we would update firewall rules to block or limit the traffic causing the issue, especially ICMP floods. We would also rely on the IDS/IPS system to automatically detect and block suspicious activity as it happens. Once the traffic is under control, we would double-check that everything is stable and that no malicious activity is still going on.

For investigation, we would look at firewall logs, IDS alerts, and network monitoring data to figure out where the attack came from and how it worked. SIEM logs would also help us see the full timeline so we can understand exactly what happened from start to finish.

To improve recovery in the future, we should make sure our backup and recovery processes are tested regularly and work smoothly. It would also help to automate more of the recovery steps so systems can be restored faster. Doing regular practice drills and reviewing our response plan would help us be better prepared next time.

Recover

To recover from this incident, the organization first needs key information like firewall logs, IDS/IPS alerts, network traffic data, and system performance records from during the attack. This helps confirm that the malicious ICMP traffic has fully stopped and also shows when normal activity began again. Having access to backup configurations of firewall rules and network settings is also important so systems can be restored quickly and correctly.

The recovery process involves gradually bringing network services back online, starting with the most critical systems to ensure business operations can continue. Firewalls must be verified and properly configured before full restoration, especially with the new ICMP filtering and IP verification rules in place. IDS/IPS systems and monitoring tools should stay active during recovery to make sure no further suspicious activity occurs.

To support recovery, the organization should rely on its backup and disaster recovery systems, which allow damaged or disabled services to be restored quickly. Regular recovery procedures, such as system snapshots and configuration backups, help speed up restoration and reduce downtime. After everything is restored, the team should review the recovery process to identify any delays or weaknesses and improve future response efforts.